

Basics of SIR Models

Some Terms

Closed v Open populations

- Closed populations have no additions due to births or immigration and no losses due to death or emigration – all dynamics due **only** to infection
 - Unrealistic, but simple and a good place to start
- Open populations can have births, immigration, deaths, emigration
 - Either explicitly or implicitly

Equilibrium and Transient

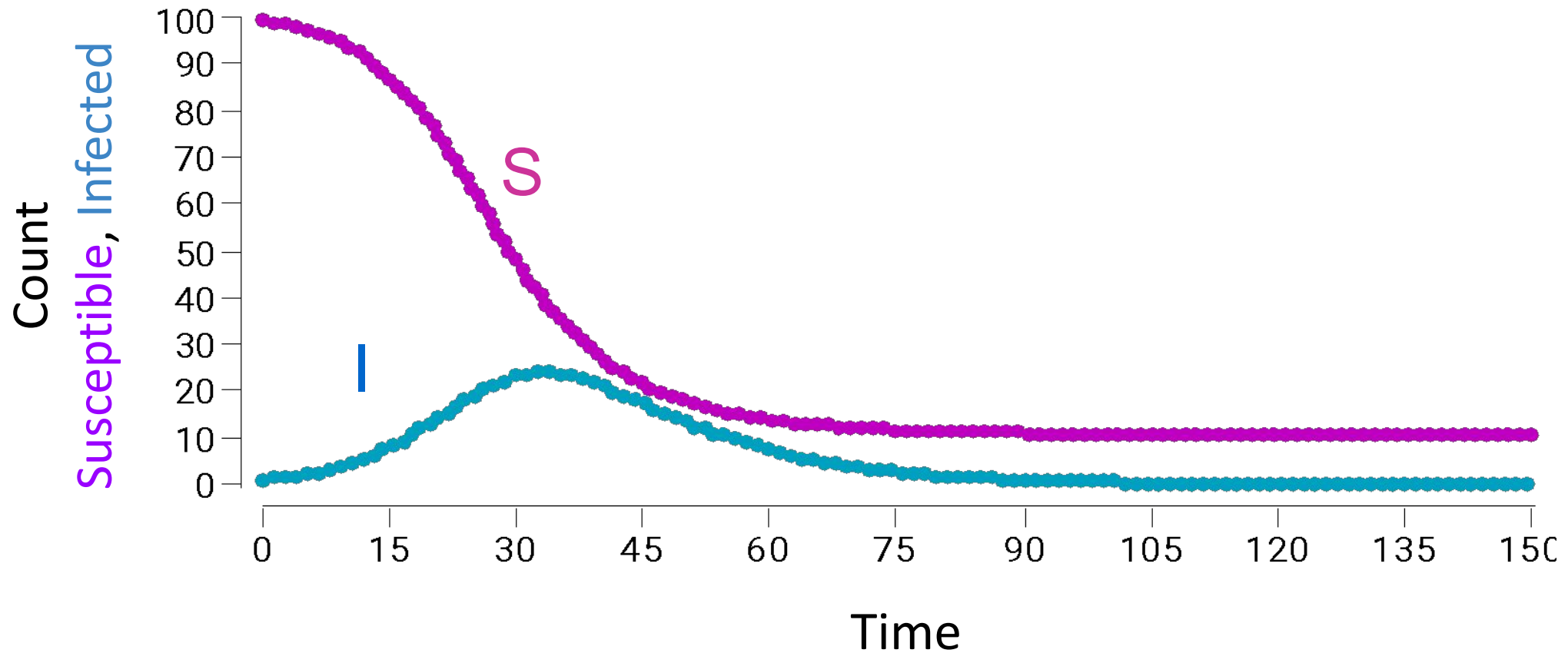
- At equilibrium, the values of all states are constant – individuals may still get sick, recover, etc, but all changes balance each other out
- Dynamics are transient if the states are continuing to change ... more nuance later

The SIR Model

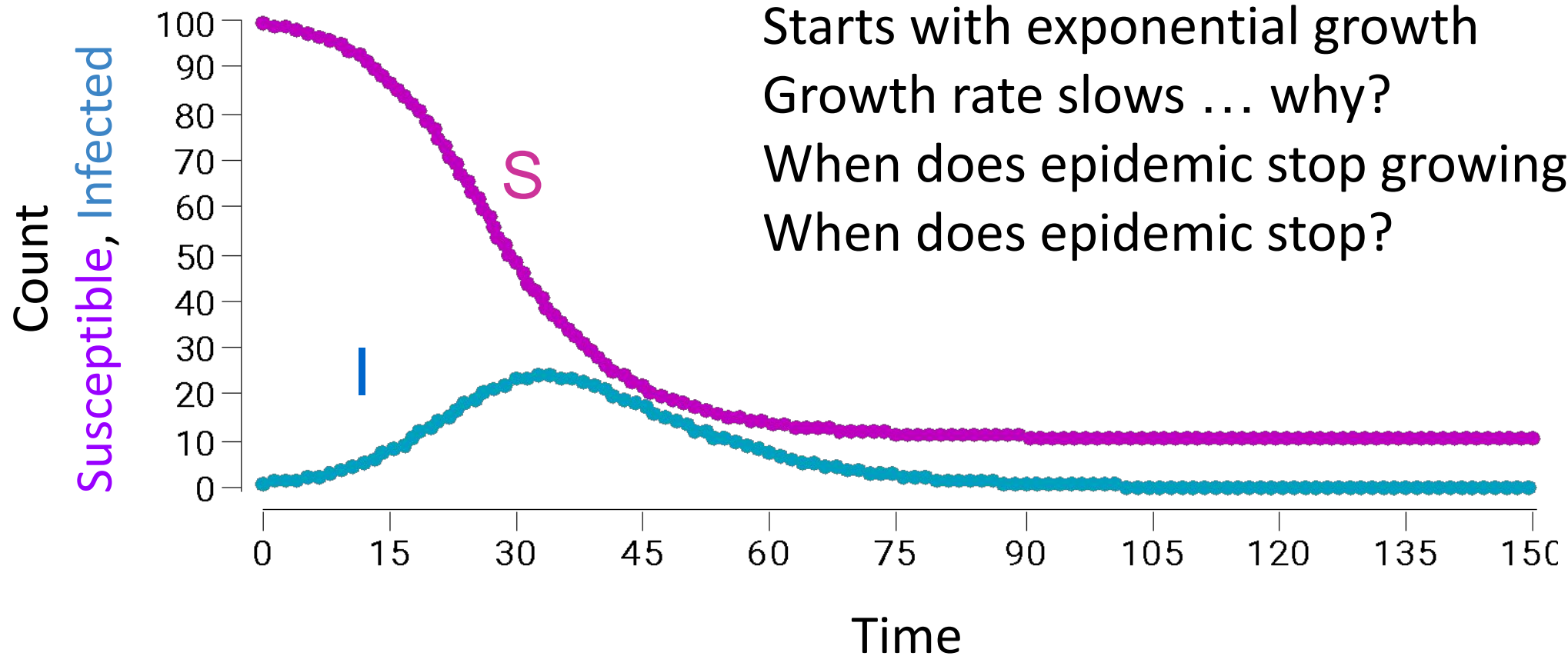
The States

S, **I**, and **R** reflect the number of individuals that are currently **susceptible**, **infected** (and infectious), and **recovered**, respectively

Idealized Epidemic in a Closed Community



Idealized Epidemic in a Closed Community



The SIR Model

$$\begin{aligned}\frac{dS}{dt} &= \\ \frac{dI}{dt} &= \\ \frac{dR}{dt} &= \end{aligned}$$

Since we're talking about how the epidemic grows, or how the **states change over time**, we write the SIR model in terms of the derivative, with respect to time, of each of the states.

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

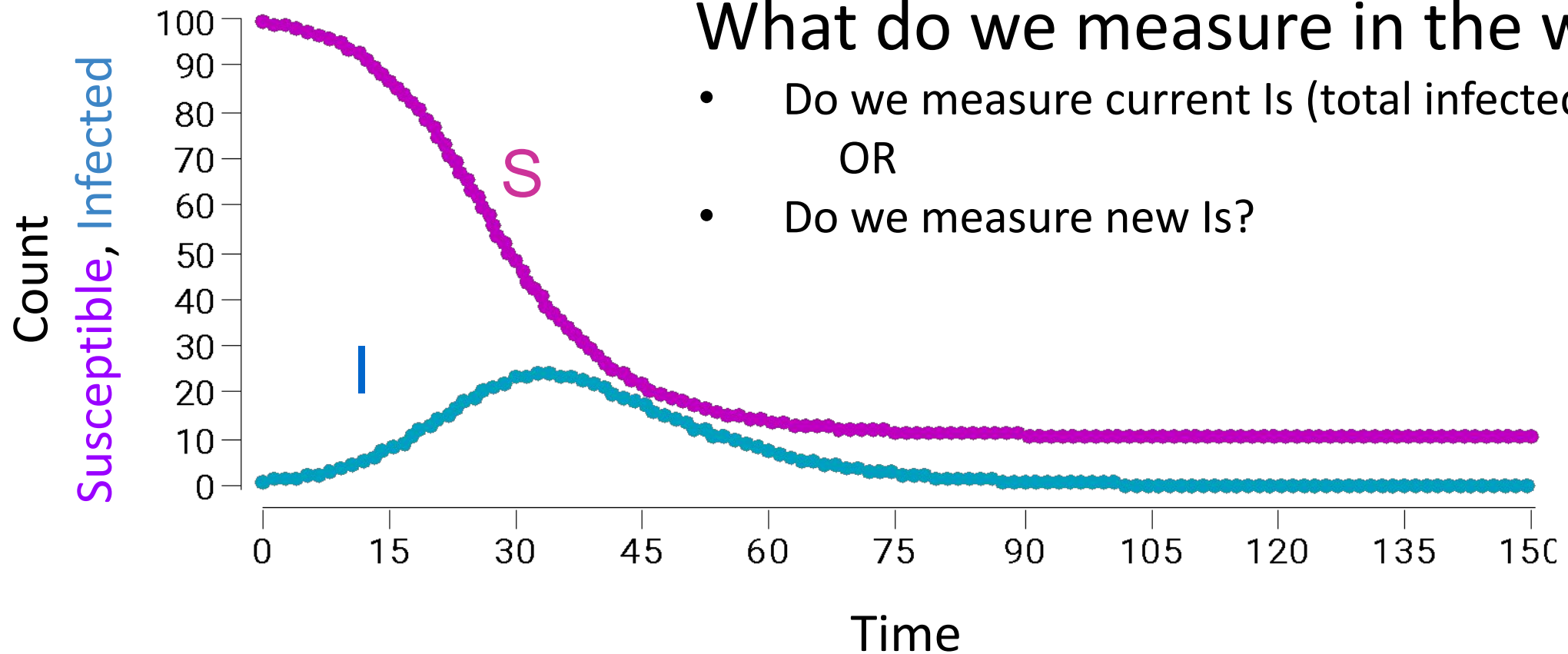
$$\frac{dR}{dt} = \beta SI + \gamma I$$

The States

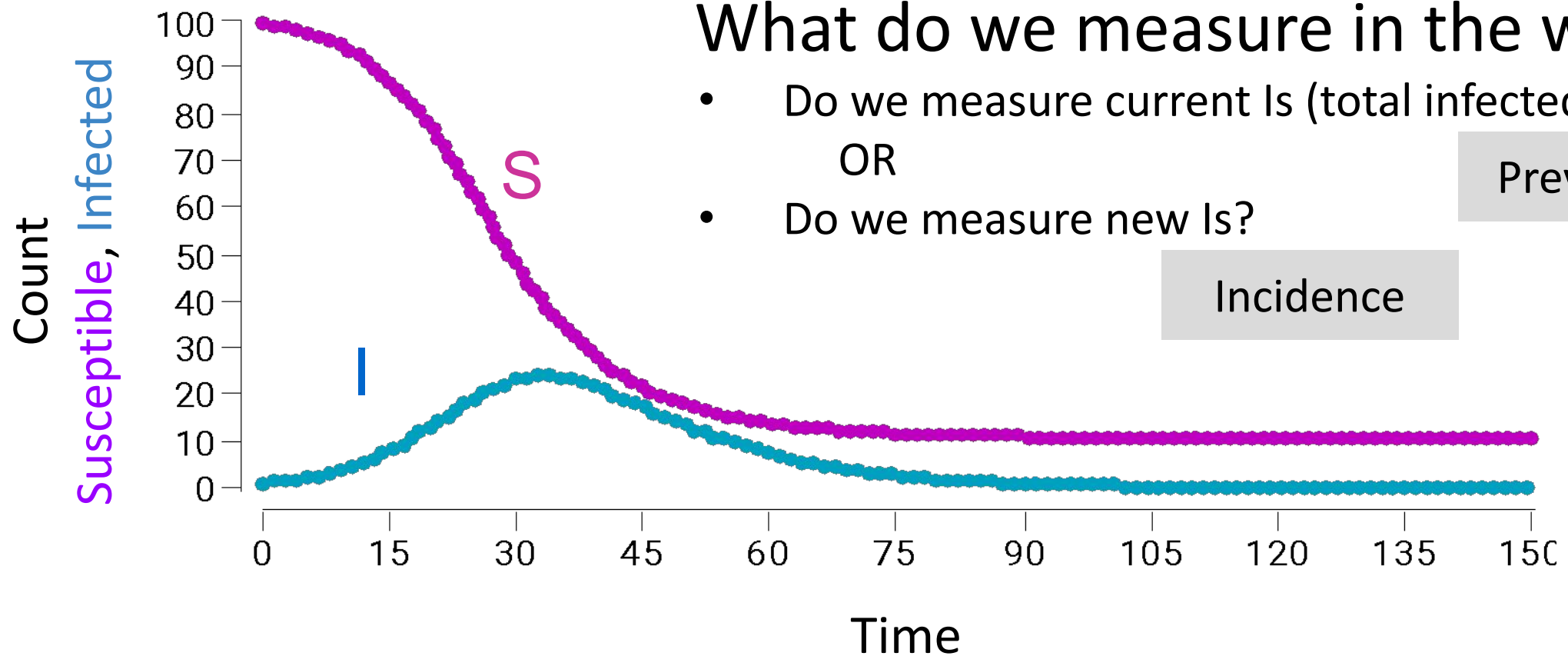
S, **I**, and **R** reflect the number of individuals that are currently **susceptible**, **infected** (and infectious), and **recovered**, respectively

Because the states are linked (an S becomes an I, and an I becomes an R) the states show up in each other's equations

Idealized Epidemic in a Closed Community



Idealized Epidemic in a Closed Community



The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

What do we measure in the world?

- Do we measure current I s (total infected),
OR
- Do we measure new I s?

How could we measure current I s?

How could we measure new I s?

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Contact Process

- This quantifies the rate at which susceptibles and infecteds interact
 - Increases with number (or proportion) of each
 - Creates non-linearity

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Contact Process

- There are lots of ways to adjust this ... the most (in)famous of which is: $S \frac{I}{N}$

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Contact Process

- There are lots of ways to adjust this ... the most (in)famous of which is: $S \frac{I}{N}$

Begon: A clarification of transmission terms in host-microparasite models

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2869860/>

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Contact Process

- What other ways might contacts change with the amount of infection?

Hint: think about behavior over the last 5 years

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Transmission Parameter

- SI defines the shape of contacts. β turns that into infectious contacts:
 - Rate of infectious contacts (not all contacts are infectious)
 - Probability of infection *given* contact

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Recovery Rate

- This is the rate, number per time, of recovery (or removal)
- If rate is high, many events happen per unit time and time between events is small

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Recovery Rate

- This is the rate, number per time, of recovery (or removal)
- So large γ means that the average duration of infection is short

The SIR Model

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = -\beta SI - \gamma I$$

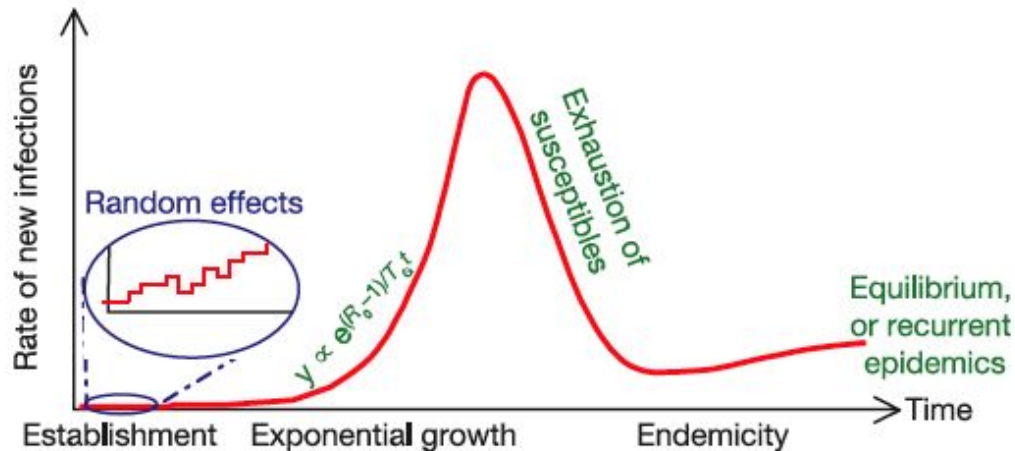
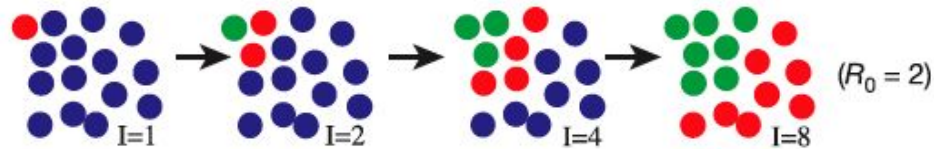
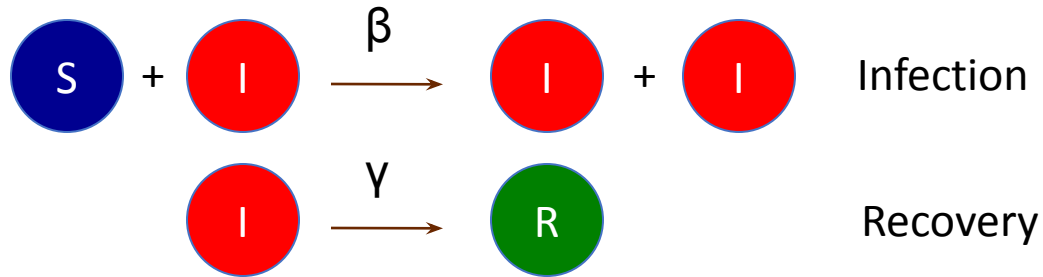
$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Recovery Rate

- This is the rate, number per time, of recovery (or removal)
- Mean duration of infection, L , is $\frac{1}{\gamma}$
if the distribution of infectious periods is exponential

How realistic is this?
Why do we do this?

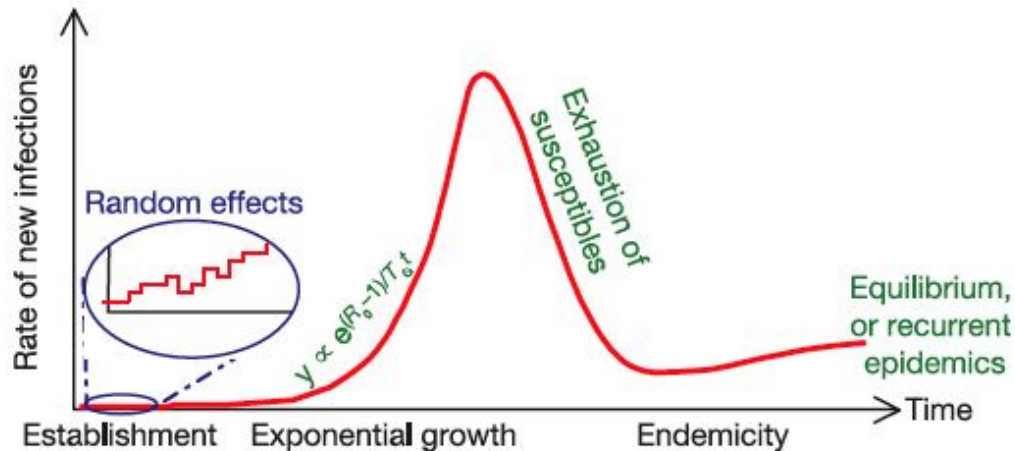
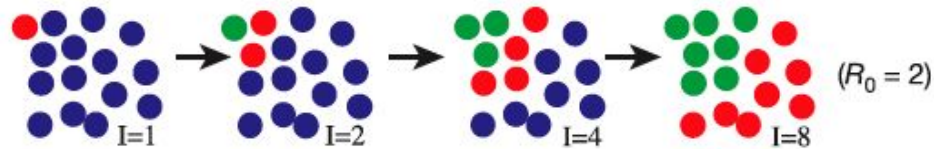
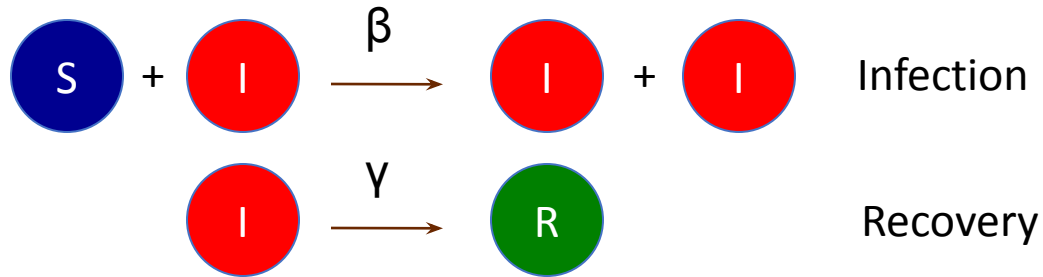
Basic Epidemic Theory



$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

At the start, when there are few infections, an epidemic grows (almost) exponentially

Basic Epidemic Theory



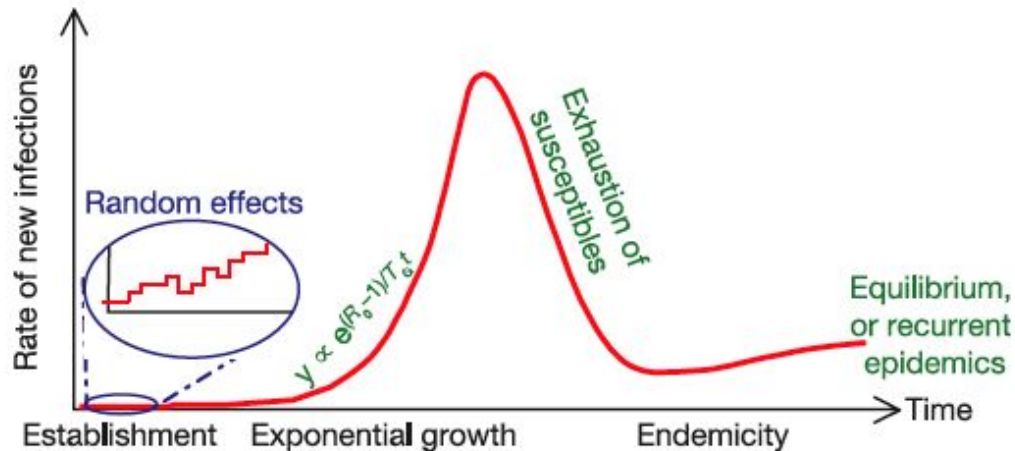
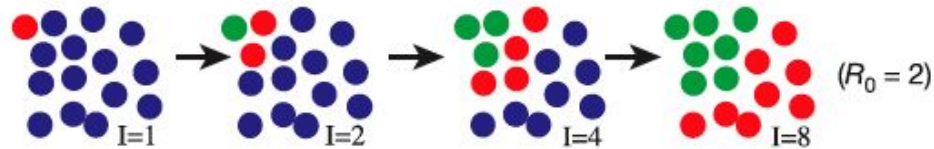
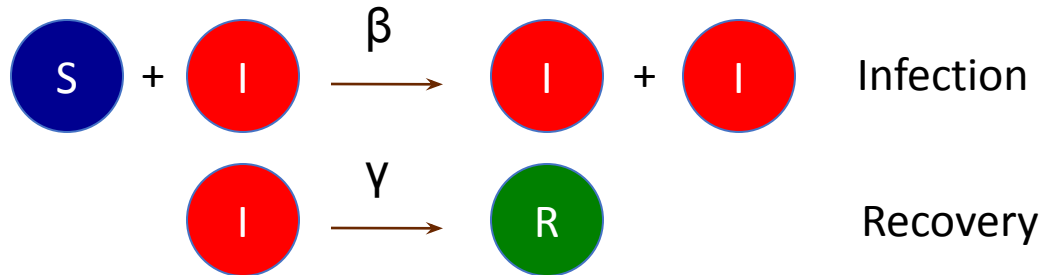
figures from Ferguson et al. *Nature* 2003

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

At the start, when there are few infections, an epidemic grows (almost) exponentially

We'll use this property later to estimate the transmission rate

Basic Epidemic Theory



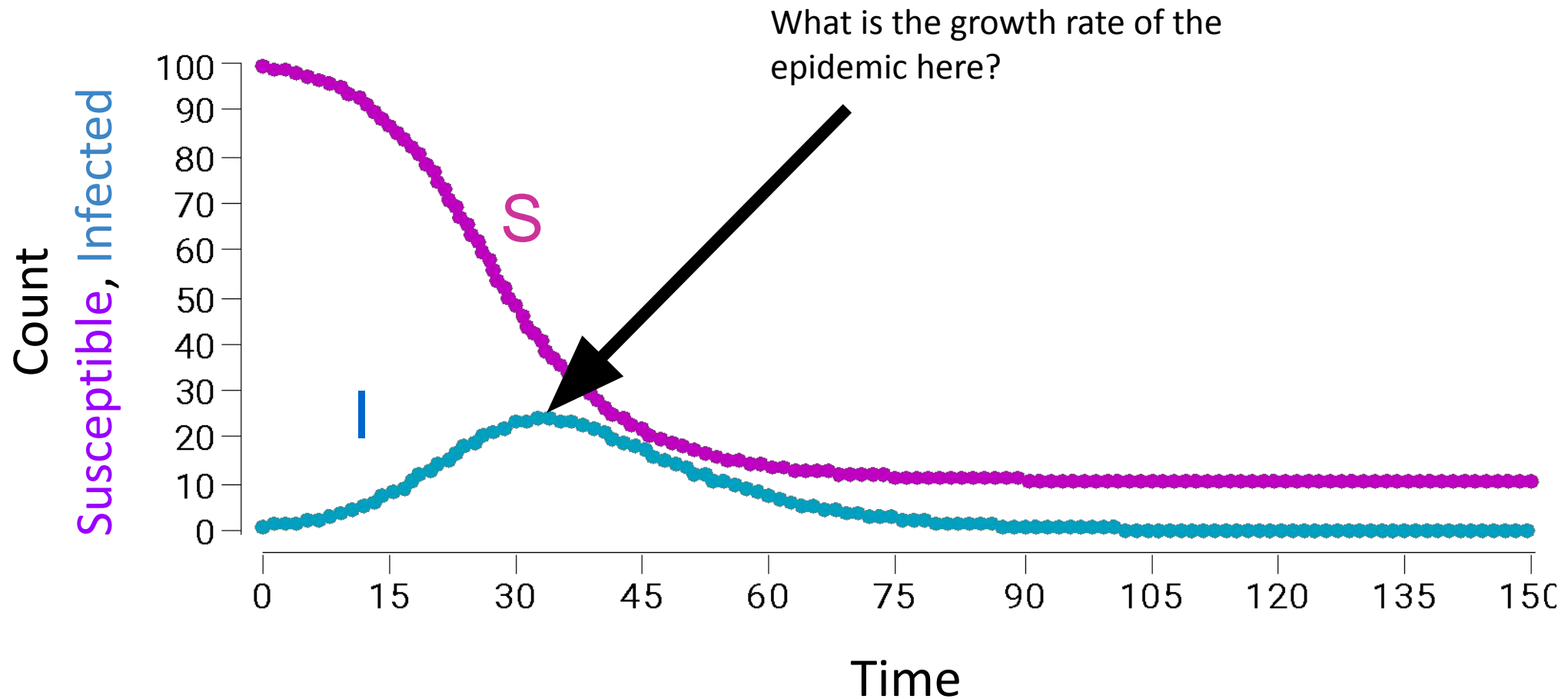
figures from Ferguson et al. *Nature* 2003

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

As individuals recover and the number susceptible declines, that growth slows because Susceptibles are being depleted

When does epidemic stop growing?

Idealized Epidemic in a Closed Community



Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 = \beta SI - \gamma I$$

$$0 = \beta S - \gamma$$

$$\beta S = \gamma$$

$$1 = \frac{\beta S}{\gamma}$$

Condition under which I
doesn't change

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 > \beta SI - \gamma I$$

$$0 > \beta S - \gamma$$

$$\beta S < \gamma$$

$$1 > \frac{\beta S}{\gamma}$$

Condition under which I declines ... epidemic fades

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 < \beta SI - \gamma I$$

$$0 < \beta S - \gamma$$

$$\beta S > \gamma$$

$$1 < \frac{\beta S}{\gamma}$$

Condition under which I
grows ... epidemic grows

Basic Reproduction Number

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = -\beta SI - \gamma I$$

Under what conditions can the infectious compartment grow?

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$0 < \beta SI - \gamma I$$

$$0 < \beta S - \gamma$$

$$\beta S > \gamma$$

$$1 < \beta SL$$

Recall that $\frac{1}{\gamma} = L$ is the
mean duration of infection

R_0 : The Basic Reproduction Number

$$R_0 = \frac{\beta S}{\gamma} = \beta SL$$

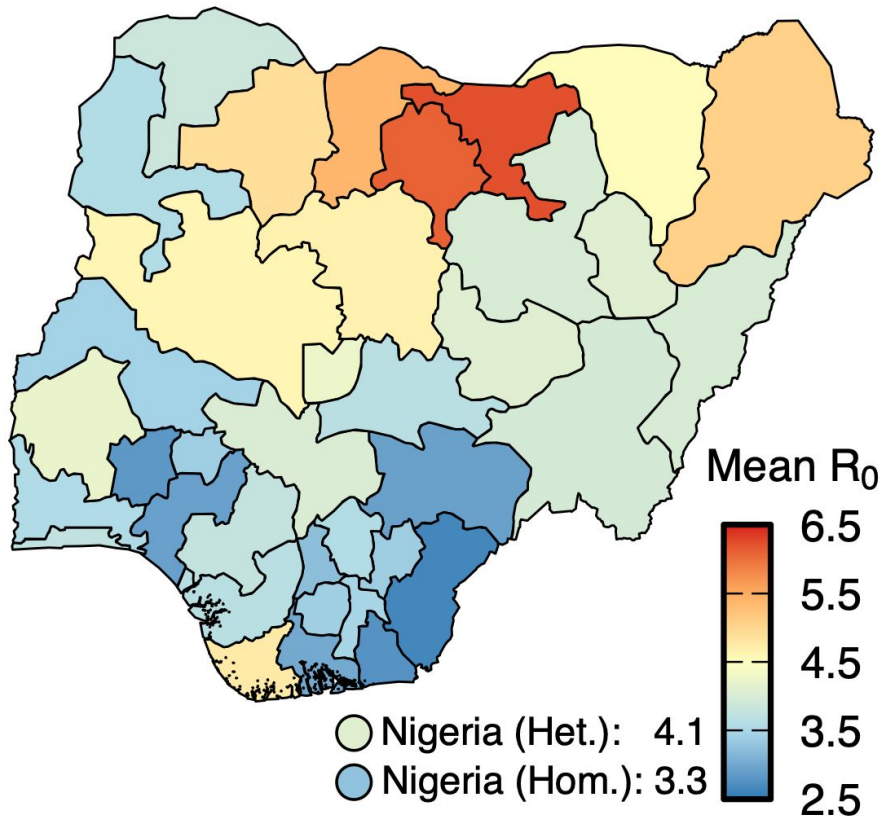
The expected number of new infections due to the first infection in a susceptible population

- A common currency
 - A function of the pathogen and the population (recall what β is)
 - Rarely observable directly
 - But closely related to many observable phenomena, as we'll see

Estimated values of R0 for various infections

Measles	England	1947	13-14
	Nigeria	1968	16-17
	Kansas	1920	5-6
Pertussis	England	1944-78	16-18
	Canada	1912	7-8
Chickenpox	USA	1912	7-8
	USA	1944	10-11

Basic Reproduction Number for Rubella



Estimated R_0 varies over 2x even within a single country

Nakase et al 2024 Vaccine

What Does This Mean for Interventions?

$$R_0 = \frac{\beta S}{\gamma} = \beta SL$$

What does this suggest for interventions?

- Reduce β
- Reduce L (increase gamma)
- Reduce S

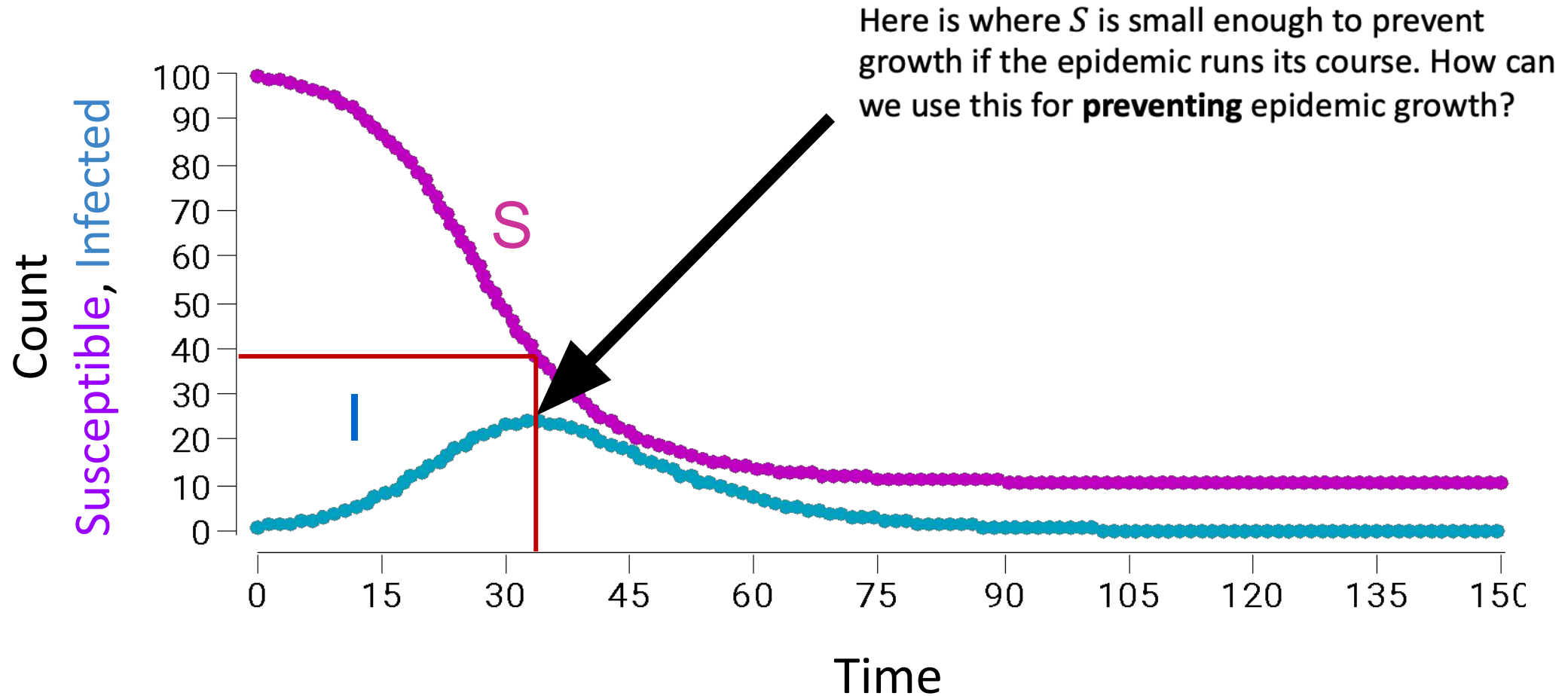
What Does This Mean for Interventions?

$$R_0 = \frac{\beta S}{\gamma} = \beta SL$$

What does this suggest for interventions?

- Reduce β
- Reduce L (increase gamma)
- Reduce S

Idealized Epidemic in a Closed Community



R_E : The Effective Reproduction Number

$$R_E = \frac{\beta p S}{\gamma} = \beta p S L$$

p is the fraction susceptible
 $1-p$ is the fraction immune

What does this suggest for interventions?

- Reduce β
- Reduce L (increase gamma)
- Reduce S

The expected number of new infections due to each infection in a population with some immunity

Herd Immunity Threshold

$$R_0 = \frac{\beta S}{\gamma} = \beta SL$$

$$R_0 = \beta SL$$

$$1 = \frac{\beta SL}{R_0}$$

$$1 = \frac{1}{R_0} S \beta L$$

What fraction of Susceptibles
need to be immune in order for

$$\frac{1}{R_0} S$$

to remain?

Herd Immunity Threshold

$$R_0 = \frac{\beta S}{\gamma} = \beta SL$$

$$R_0 = \beta SL$$

$$1 = \frac{\beta SL}{R_0}$$

$$1 = \frac{1}{R_0} S \beta L$$

What fraction of Susceptibles
need to be immune in order for

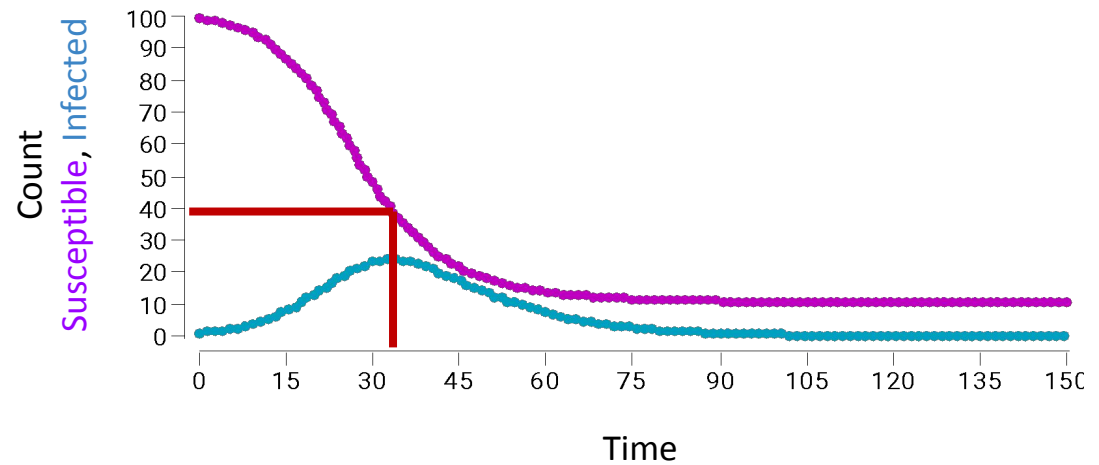
$$\frac{1}{R_0} S$$

to remain?

$$T_c = 1 - \frac{1}{R_0}$$

Herd Immunity Threshold

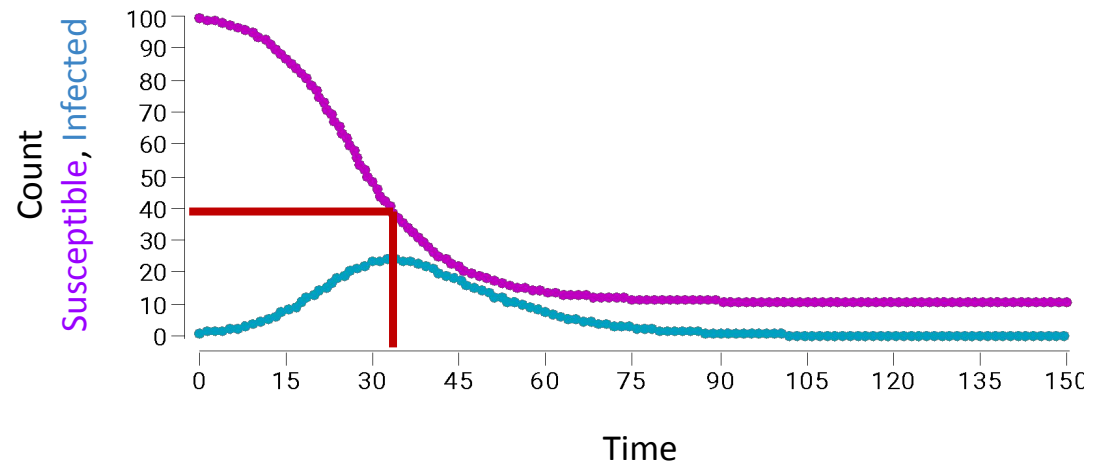
If $T_c = 1 - \frac{1}{R_0}$ are immune ***before*** an outbreak then it won't be able to grow (on average)



If an outbreak takes off it ***WILL NOT*** stop when $T_c = 1 - \frac{1}{R_0}$ are immune

Herd Immunity Threshold

If $T_c = 1 - \frac{1}{R_0}$ are immune ***before*** an outbreak then it won't be able to grow (on average)



If an outbreak takes off it ***WILL NOT*** stop when $T_c = 1 - \frac{1}{R_0}$ are immune

Why not?

Final Size Calculation

$$R_{\infty} = 1 - e^{-R_0 R_{\infty}}$$

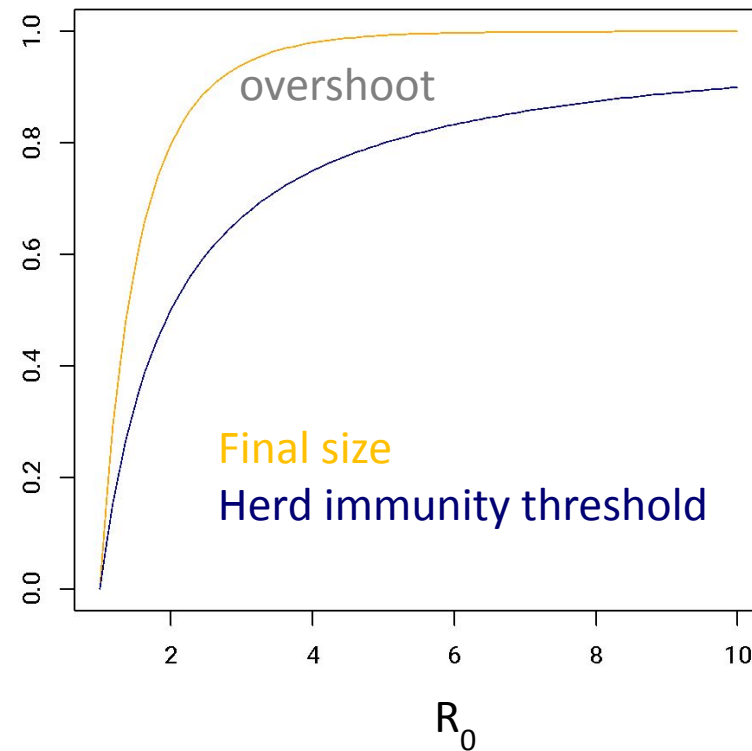
Where R_{∞} is the proportion of the population infected at the end of the epidemic (the proportion in the R class at the end)

Citation: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3506030/>

Comparing T_c and Final Size

Many more individuals will become infected in an epidemic (on average) than need to be immunized **BEFORE** an epidemic

Herd Immunity is a relevant concept throughout an epidemic (and helps stop them), the Herd Immunity Threshold is only relevant for preventing, not stopping outbreaks.



What Happens When Births Are Added?

$$\frac{dS}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta S$$

$$\frac{dI}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta I$$

$$\frac{dR}{dt} = \delta N - \beta SI - \gamma I - \alpha R - \delta R$$

δ is birth and death rate

α is disease induced death rate

What Happens When Births Are Added?

$$\frac{dS}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta S$$

$$\frac{dI}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta I \quad \text{When there's NO infection?}$$

$$\frac{dR}{dt} = \delta N - \beta SI - \gamma I - \alpha R - \delta R$$

δ is birth and death rate

α is disease induced death rate

What Happens When Births Are Added?

$$\frac{dS}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta S$$

$$\frac{dI}{dt} = \delta N - \beta SI - \gamma I - \alpha I - \delta I$$

$$\frac{dR}{dt} = \delta N - \beta SI - \gamma I - \alpha R - \delta R$$

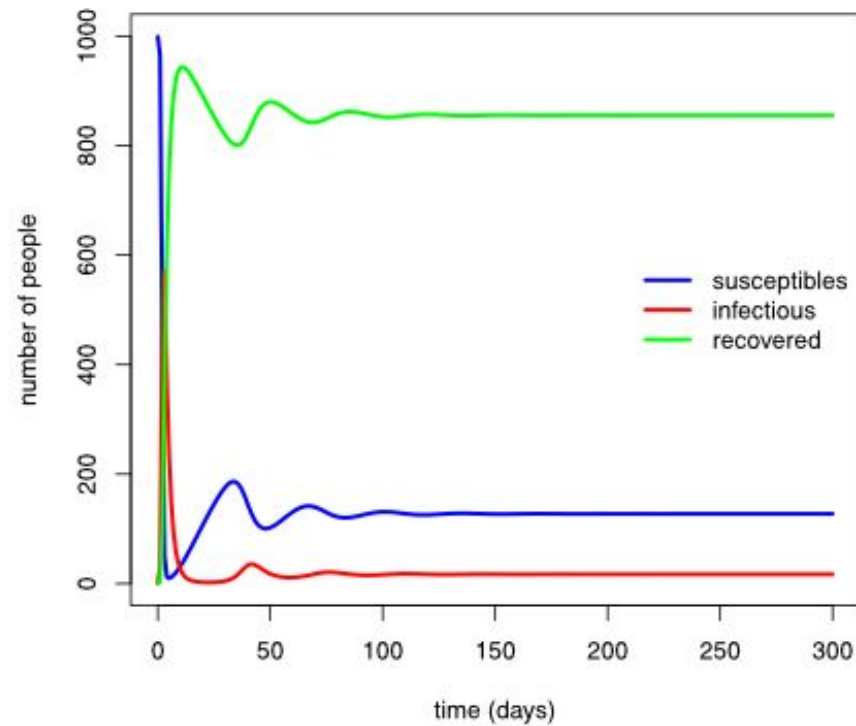
When there's very little infection?

δ is birth and death rate

α is disease induced death rate

Dynamics Over Time

Note that after initial overshoot, susceptibles build back up until a new outbreak occurs, the second is smaller, and the third smaller than that



Some Terms

Closed v Open populations

- Closed populations have no additions due to births or immigration and no losses due to death or emigration – all dynamics due **only** to infection
 - Unrealistic, but simple and a good place to start
- Open populations can have births, immigration, deaths, emigration
 - Either explicitly or implicitly

Equilibrium and Transient

- At equilibrium, the values of all states are constant – individuals may still get sick, recover, etc, but all changes balance each other out
- Dynamics are transient if the states are continuing to change ... more nuance later (really)

When Does I Stop Growing?

$$\frac{dI}{dt} = \beta SI - \gamma I - \alpha I - \delta I$$

$$0 < \beta SI - \gamma I - \alpha I - \delta I$$

$$\beta SI > \gamma I + \alpha I + \delta I$$

$$\beta S > \gamma + \alpha + \delta$$

$$1 < \frac{\beta S}{\gamma + \alpha + \delta} \equiv R_0$$

δ is birth and death rate

α is disease induced death rate

When Does I Stop Growing?

$$\frac{dI}{dt} = \beta SI - \gamma I - \alpha I - \delta I$$

$$0 < \beta SI - \gamma I - \alpha I - \delta I$$

$$\beta SI > \gamma I + \alpha I + \delta I$$

$$\beta S > \gamma + \alpha + \delta$$

$$1 < \frac{\beta S}{\gamma + \alpha + \delta} \equiv R_0$$

Why do these terms show up
in equation for R_0 ?

δ is birth and death rate

α is disease induced death rate

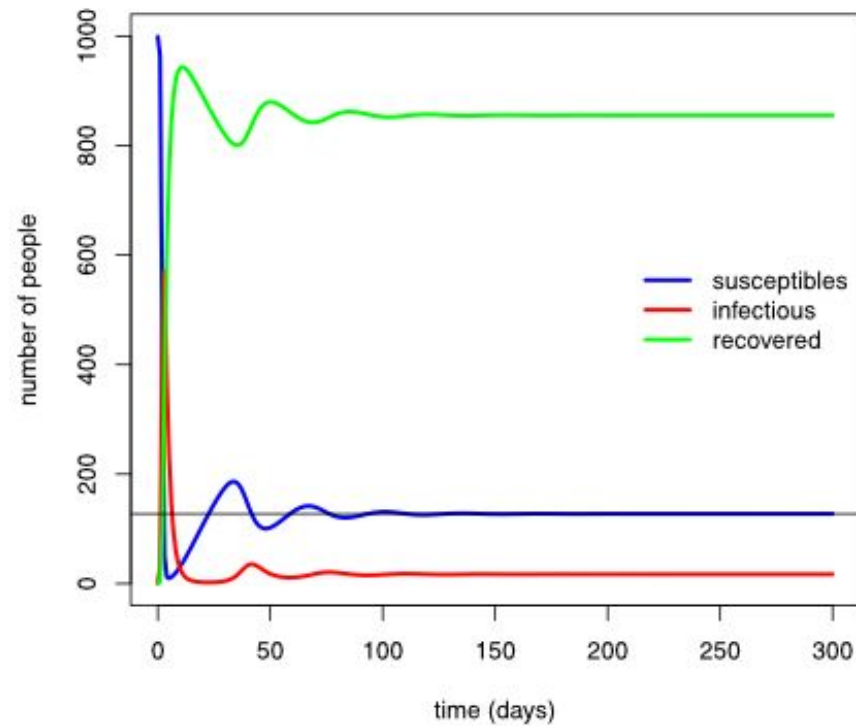
Equilibrium Dynamics

The stable equilibrium proportion susceptible is

$$\sim \frac{1}{R_0}$$

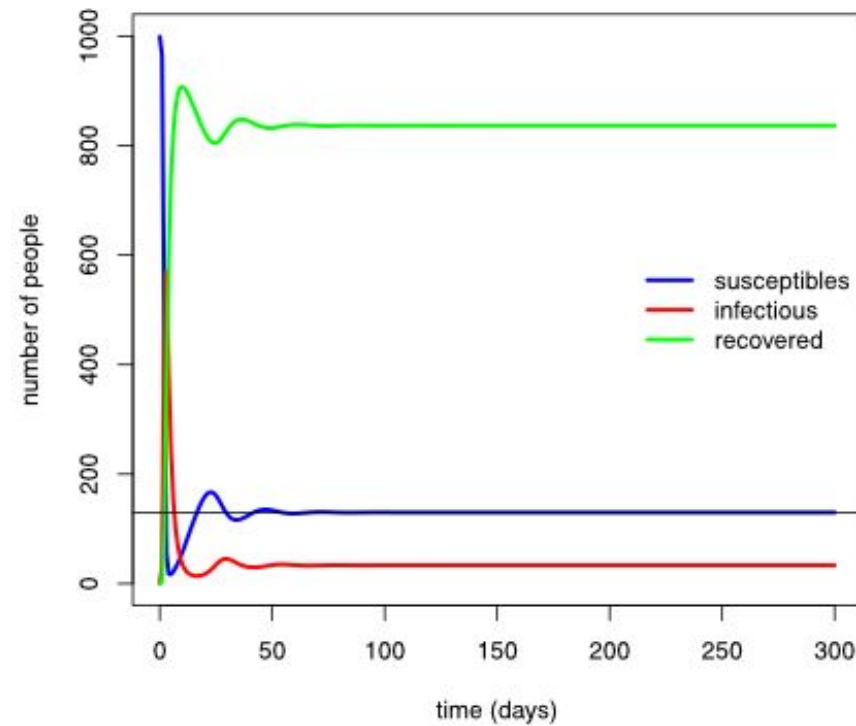
and the stable proportion recovered (immune) is

$$\sim 1 - \frac{1}{R_0}$$



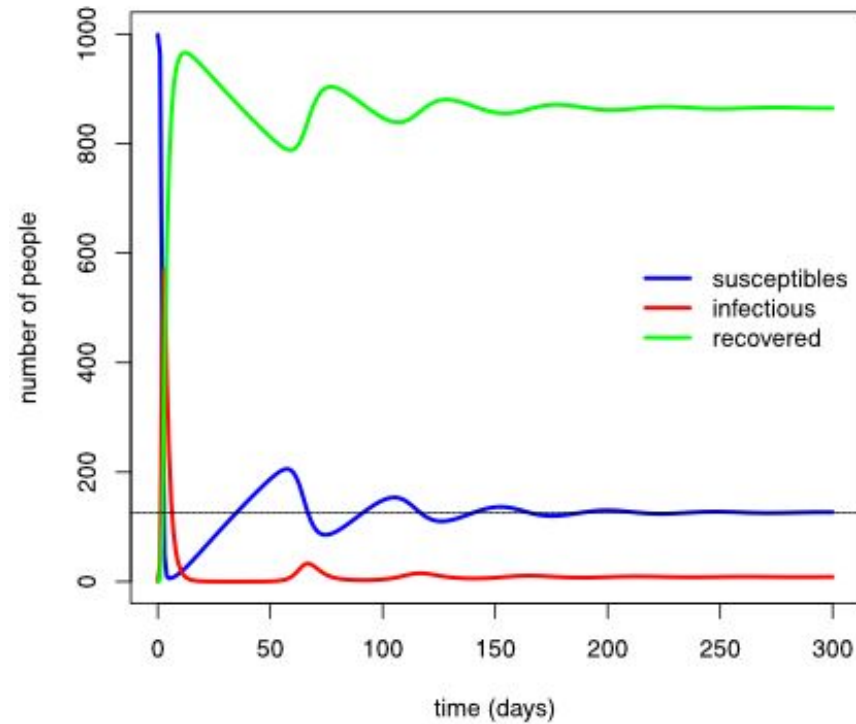
Birth Rate Changes the Speed to Equilibrium

If we increase the birth rate, it takes less time to reach equilibrium under the assumption that the population isn't growing



Birth Rate Changes the Speed to Equilibrium

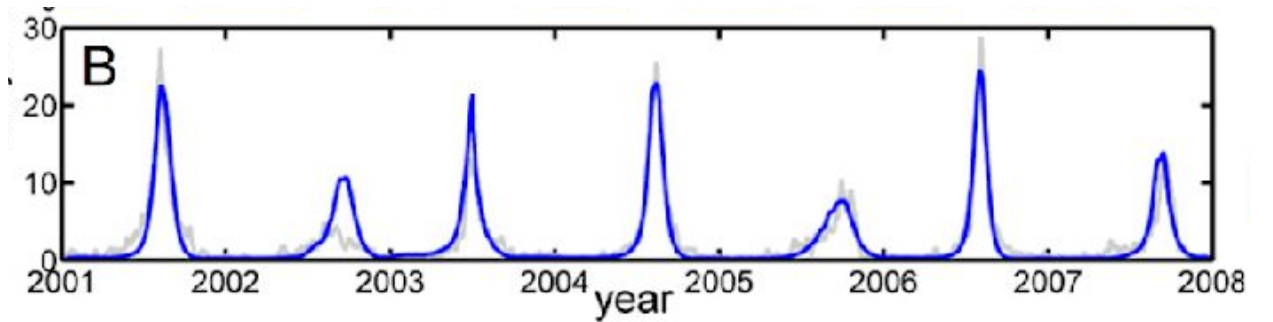
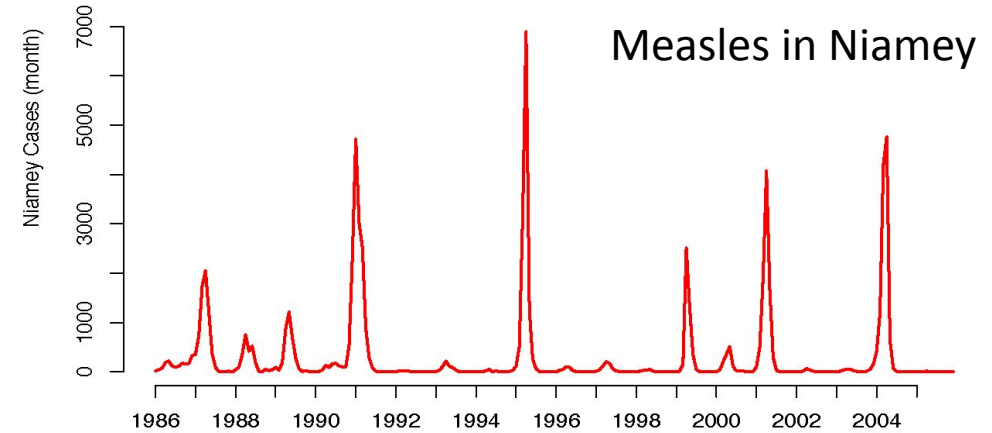
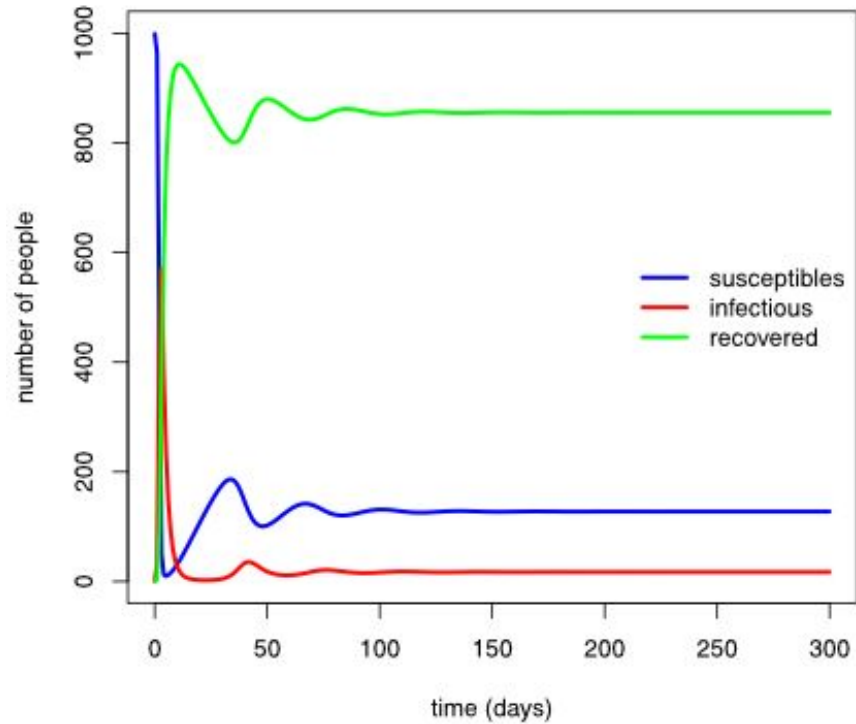
If we decrease the birth rate,
it takes longer to reach
equilibrium under the
assumption that the
population isn't growing



What about growing populations?

- Growing populations have more susceptibles added than recovered being taken away (by death)
 - So a greater fraction susceptible, less indirect protection, and more transmission
- More of those susceptibles are young, so if young and old have different contact rates, then transmission and dynamics will differ in young vs. old populations ...

Seasonality and Cycles



Influenza in Jerusalem

Some Terms

Closed v Open populations

- Closed populations have no additions due to births or immigration and no losses due to death or emigration – all dynamics due **only** to infection
 - Unrealistic, but simple and a good place to start
- Open populations can have births, immigration, deaths, emigration
 - Either explicitly or implicitly

Equilibrium and Transient

- At equilibrium, the values of all states are constant – individuals may still get sick, recover, etc, but all changes balance each other out
- An *attractor* is collection of states towards which a system tends – it's regular and predictable, but not static.
- Dynamics are transient if they are neither of these ... which is most of time

Seasonality and Cycles

- Long term dynamics often exhibit regular fluctuations around the equilibrium levels because of seasonal changes in

- Environmental conditions
- Behavior
- Population movement/aggregation
- Vector seasonality

Examples

Influenza
Lassa fever
Legionellosis
Leptospirosis
Meningococcal meningitis
Polio
Typhoid

Seasonality and Cycles

- Long term dynamics often exhibit regular fluctuations around the equilibrium levels because of seasonal changes in

- Environmental conditions
- Behavior
- Population movement/aggregation
- Vector seasonality

Examples

Chickenpox

Measles

Pertussis

Rubella

Seasonality and Cycles

- Long term dynamics often exhibit regular fluctuations around the equilibrium levels because of seasonal changes in

- Environmental conditions
- Behavior
- Population movement/aggregation
- Vector seasonality

Examples

Influenza
Lassa fever
Legionellosis
Leptospirosis
Meningococcal meningitis
Polio
Typhoid

Examples

Chickenpox
Measles
Pertussis
Rubella

Modeled as a temporal change in β

Seasonality and Cycles

- Long term dynamics often exhibit regular fluctuations around the equilibrium levels because of seasonal changes in

- Environmental conditions
- Behavior
- Population movement/aggregation
- Vector seasonality

Examples

Measles

Meningococcal meningitis

Modeled as a temporal change in β *or* S

Seasonality and Cycles

- Long term dynamics often exhibit regular fluctuations around the equilibrium levels because of seasonal changes in

- Environmental conditions
- Behavior
- Population movement/aggregation
- Vector seasonality

Examples

Chikungunya

Dengue

Malaria

Trypanosomiasis

West Nile Virus

Yellow Fever

Requires a new compartment for the vector populations